ChE 301 – Chemical Engineering Thermodynamics –Spring 2020
Student Learning Outcome 1

Rubric for assessment problem on final exam (Problems #1,2,4,5)

1. **Tank Pressurization Calculations with Redlich-Kwong Equation of State.**
2. **Air Dew Point Calculations**
3. VLE Data Reduction with the Peng-Robinson Equation of State
4. **Excess Enthalpy Models for Isopropanol-Water Mixtures**
5. **VLE Data Reduction with the NRTL Model**

SLO 1 Performance Indicator (PI)	**(1)** Unsatisfactory	**(2)** Developing	**(3)** Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject **Problem 1A,1B** **100 pts**	Does not understand the connection between mathematical models and the system or process to be analyzed or designed **(<50 pts)**	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development **(50-70 pts)**	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering **(>70 pts)**
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra **Problem 4A,4B,4C** **100 pts**	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems **(<50 pts)**	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving **(50-70 pts)**	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems **(>70 pts)**
4. Translates academic theory into engineering applications **Problem 2,5A,5B,5C** **100 pts**	Does not appear to grasp the connection between theory and the problem **(<50 pts)**	Some gaps in understanding the application of theory to the problem and expects theory to predict reality **(50-70 pts)**	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality **(>70 pts)**

RESULTS BY PERFORMANCE INDICATOR			
Rubric #	1	2	4
Mapped SO	1	1	1
Average	2.56	2.78	2.44
Std Dev	0.78	0.65	0.86

Average scores were fairly uniform although there was some variation among students. More students were strong in solving engineering problems using theoretical thermodynamic concepts , and concepts of algebra, calculus, differential equations and/or linear algebra. Overall performance was satisfactory for all performance indicators.

Problem 1: Tank Pressurization Calculations with the Redlich-Kwong EOS

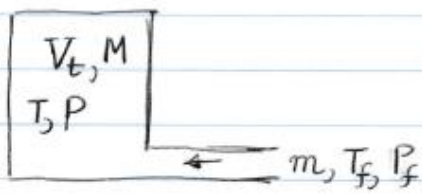
A cylindrical tank with volume $\left\{V = 25\, m^3\right\}$ is originally filled with nitrogen at conditions of $\left\{T_o = 25\,^0C \quad \text{and} \quad P_o = 1 bar\right\}$. The tank must be pressurized to a pressure of $\left\{P = 25 bar\right\}$ using a high pressure line which supplies nitrogen at a constant temperature and pressure equal to $\left\{T = 25\,^0C \quad \text{and} \quad P = 50\, bar\right\}$.

A. Prove that by combining the dynamic energy and mass balances we obtain the equation: $\boxed{d\left(MU\right) = H\, dM}$ where M is the mass(in moles) of nitrogen inside the tank at any time, U is the molar internal energy of nitrogen inside the tank and H is the molar enthalpy of the nitrogen in the high pressure line feeding the tank

B. Integrate the equation obtained above and calculate the amount of nitrogen added to the tank during the pressurization process using (a) the ideal gas law and (b) the Redlich-Kwong equation of state. What is the error in assuming that the gas is ideal.

Problem 1. Tank Pressurization Calculations with RK EOS

Consider the following pictorial for the Pressurization Process



V_t = tank volume = $25 m^3$
T, P = temperature & pressure @ any time
T_f, P_f = temperature & pressure of feed line
m = mass flow rate in, kg/s
M = mass of Nitrogen in tank @ any time

(A) Mass Balance

$$\frac{dM}{dt} = m \qquad (1)$$

Energy Balance

Ignoring Potential & Kinetic Energy Contributions we obtain

$$\frac{d(MU)}{dt} = mH_f \qquad (2)$$

where U [=] Specific Mass Internal Energy, J/kg
H_f [=] Specific Mass Enthalpy of Feed line, J/kg

Combining (1) & (2) we obtain

$$\frac{d(MU)}{dt} = \frac{dM}{dt} H_f \Rightarrow \boxed{d(MU) = H_f \, dM} \quad (3)$$

(B) Integrating Eq. (3) we obtain $MU = M_o U_o + (M - M_o) H_f$ (4)
Recognizing that $H = U + PV \Rightarrow U = H - PV = H - P/\rho$
and therefore equation (4) becomes

$$M(H - H_f - P/\rho) = M_o(H_o - P_o/\rho_o - H_f) \qquad (5)$$

where M_o = initial mass of N_2 in tank

ρ_o = initial density of N_2 in tank

P_o = initial pressure of N_2 in tank

T_o = initial temperature of N_2 in tank

Also we have $M = \rho V_t$ and $M_o = \rho_o V_t$ and therefor equation (5) becomes

$$\boxed{\rho\left(H - H_f - \frac{P}{\rho}\right) = \rho_o\left(H_o - \frac{P_o}{\rho_o} - H_f\right) \quad (6)}$$

Equation (6) can be solved by trial & Error (or using Goal Seek in Excel). The process is described in detail in the steps below:

1. Guess a temperature T^o = (say 50°C)
2. Using <u>CHE301 Calculator</u> Calculate
   a. Compressibility factor
   b. Enthalpy, H
   c. Density, ρ
3. Substitute values in Equation (6)
4. Repeat [Goal Seek] till equation (6) is satisfied

The steps apply both for <u>IDEAL GAS</u> and for gas following any equation of state. Final Resuets are shown in the Excel image depicted below.

NUMERICAL SOLUTION

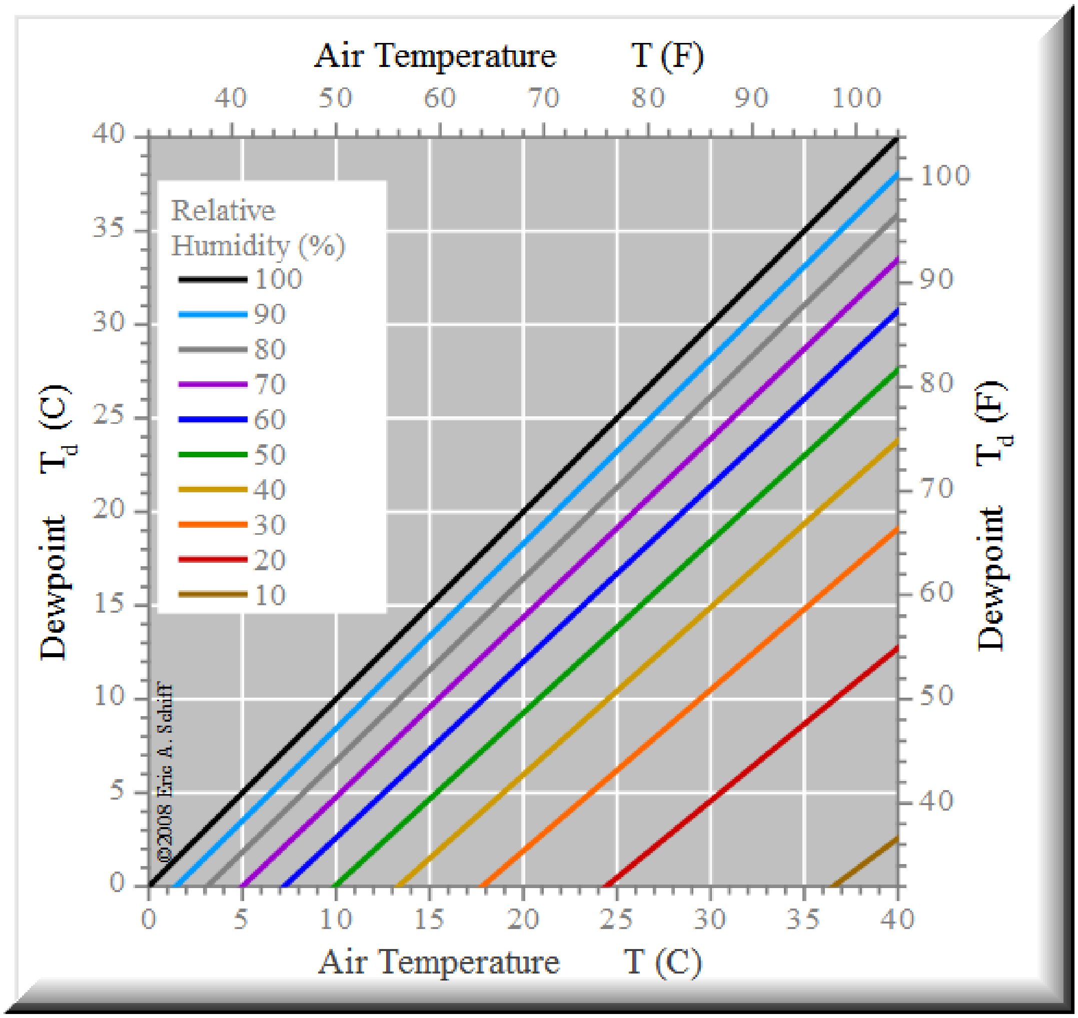
NOTEBOOK

	IDEAL	RK EOS
Feed Line Temperature, C	25	25
Feed Line Pressure, bar	50	50
Feed Line Compressibility Z	1.0000	0.9874
Feed Line Enthalpy, kJ/kmol	0.0000	-300.3600
Initial Temperature, C	25	25
Initial Pressure, bar	1	1
Initial Compressibility Z	1	0.9996
Initial Density, kmol/m3	0.040339	0.040355
Initial Enthalpy, kJ/kmol	0.0000	-6.4120
Final Temperature	137.14	127.82
Final Pressure	25	25
Final Compressibility Z	1	1.0033
Final Density	0.732838	0.747393
Final Enthalpy	3274.9416	2926.6726
Equation (See Below)	0.00000	0.00000
Tank Volume, m3	25	25
Initial Mass, kmol	1.0085	1.0089
Final Mass, kmol	18.3209	18.6848
Mass Added, kmol	**17.3125**	**17.6760**

$$\rho\left(H-H_{\text{Feed}}-\frac{P}{\rho}\right)-\rho_o\left(H_o-H_{\text{Feed}}-\frac{P_o}{\rho_o}\right)=0$$

Problem 2: **Dew Point Calculations**

On Tuesday July 8, 2003 in Dhahran Saudi Arabia, the recorded temperature at 12:00 noon was 104ºF and the sea level pressure 29.39 inHg. The relative humidity was recorded at 72%. Estimate the dew point of the air from the chart given below, and verify your estimation using a thermodynamic model of your choice. If you decide to use Henry's law for oxygen, nitrogen, argon and carbon dioxide values for the Henry's law constants for these gases can be found in your notes. The composition of dry air can be found in the open literature.



Problem 2. Dew Point Calculations

We begin by stating the composition of Dry air. The data are taken from Mackenzie, F.T. and J.A. Mackenzie (1995) Our Changing Planet, Prentice-Hall, Upper Saddle River, NJ pp. 288-307

N_2 78.084 % mol

O_2 20.947 % mol

Argon 0.934 % mol

CO_2 0.035 % mol

Trace amounts are added into Argon. Notice that the composition of CO_2 might be higher today.

We now must calculate the composition of the humid air. From the definition of Relative Humidity

$$RH = \frac{P_{H_2O}}{P^{sat}_{H_2O}} \Rightarrow RH = \frac{y_{H_2O} P}{P^{sat}_{H_2O}}$$

At $T = 104\ °F$ and $P = 29.39$ in Hg and $RH = 0.72$ the $P^{sat}_{H_2O}$ from the Steam Properties of the CHE301 Calculator we obtain $P^{sat}_{H_2O} = 1.07$ psia $= 2.178542$ in Hg and therefore

$$y_{H_2O} = \frac{(RH)\, P^{sat}_{H_2O}}{P} = \frac{(0.72)(2.178542)}{29.39} \Rightarrow \boxed{y_{H_2O} = 0.0534}$$

The composition of all the components in the humid air is then $y_i = (1 - y_{H_2O})\, y_i^{DRY}$, $i = N_2, O_2, Ar, CO_2$

The composition of the humid air obtained therefore is

1.	$N_2 = 0.73917$	$T_c = -146.95\ °C$: Supercritical
2.	$O_2 = 0.19829$	$T_c = -118.55\ °C$: Supercritical
3.	$Ar = 0.00884$	$T_c = -122.35\ °C$: Supercritical
4.	$CO_2 = 0.00033$	$T_c = 31.05$
5.	$H_2O = 0.05337$	$T_c = 374.15$

Assuming that the air behaves as ideal gas and that the composition of the gases in the liquid can be calculated by Raoult's law and Henry's law for the supercritical components as shown in the table above, the dew point equation becomes

$$\sum \frac{z_i}{K_i} = 1 \Rightarrow \frac{z_1 P}{H_1} + \frac{z_2 P}{H_2} + \frac{z_3 P}{H_3} + \frac{z_4 P}{H_4} + \frac{z_5 P}{P_{H_2O}^{sat}} = 1$$

where for CO_2 we have also used Henry's law because the critical temperature of CO_2 is close to the dew point. The values of H_1, H_2, H_3 and H_4 are $H_1 = 87{,}650$ bar, $H_2 = 44{,}380$ bar, $H_3 = 40{,}000$ bar and $H_4 = 1{,}670$ bar. We use the calculator to get the dew point by using the following steps

Step 1. Enter Components and molar composition

Step 2. Enter Pressure and a guess for dew Temperature $T^o = 30°C$

Step 3. Calculate P^{sat} for water and enter Henry's constants for all other components

Step 4. Calculate the K-values

Step 5. Set the value of $\alpha = 1$

Step 6. Solve for the DewPoint temperature using Goal seek

For Goalseek Set Cell: Rachford-Rice Equation

To Value: 0.0

By changing cell: Dew Point Temperature

An image of the Excel Calculations with the Set cell in **blue** and the changing cell in **red** is shown below:

Fluid Conditions	Mixture	N2	O2	Ar	CO2	H2O
Molar Composition	1.0000	0.73917	0.19829	0.00884	0.00033	0.05337
Temperature, C	33.7690					
Pressure, bar	0.9953					
Saturated Vapor Pressure, bar		87650.00000	44380.00000	40000.00000	1670.00000	0.05312
Saturated Vapor Temperature, C						
K-Values		88067.47240	44591.37964	40190.51793	1677.95412	0.05337
New K-Values						
Vapor to Feed Molar Ratio [a=V/F]	1.0000					
The Rachford-Rice Equation	0.0000					
Discriminant of the Cubic EOS						

Problem 4: Excess Enthalpy Model for Isopropanol-Water Mixtures

Your first project at the XYZ corporation is to develop a methodology for the calculation of the molar enthalpy of isopropyl alcohol(1)-water(2) liquid mixtures at different temperatures and pressure equal to 1 atm. There is a large number of the excess enthalpy experimental data available at different temperatures and mixture compositions and these are given in the table below: (all data are atmospheric pressure)

T=273.15K		T=308.15K		T=328.15K	
x_1	H^E, kJ/kmol	x_1	H^E, kJ/kmol	x_1	H^E, kJ/kmol
0.001	-20.934	0.0029	33.494	0.0047	-36.006
0.005	-97.134	0.013	-121.417	0.019	-115.137
0.014	-257.907	0.057	-355.878	0.063	-209.34
0.026	-439.614	0.103	-456.361	0.084	-234.461
0.083	-866.667	0.151	-439.614	0.1	-247.021
0.087	-870.854	0.351	-171.659	0.138	-198.873
0.135	-983.898	0.448	-33.494	0.158	-173.333
0.179	-921.096	0.577	157.424	0.163	-160.354
0.226	-862.481	0.673	211.015	0.25	-8.792
0.509	-401.933	0.767	217.714	0.359	146.538
0.704	-106.345	0.891	146.538	0.456	292.657
0.788	-9.211	0.929	103.833	0.615	410.306
0.864	33.494	0.969	48.148	0.637	418.68
0.904	30.982			0.789	319.871
0.939	22.609			0.903	185.057
				0.909	174.59
				0.967	69.92

A. Develop a model for the Excess Enthalpy H^E using the thermodynamic principles you have learnt in class and estimate the parameters of the model from the experimental data.
B. Using the model you developed for the excess enthalpy calculate the amount of heat in kJ per kmol of solution exchanged with the surroundings when 20 moles of isopropyl alcohol are mixed **isothermally** with 80 moles of water at 315.15 ^oK.
C. If the pure components are initially at 315.15 ^oK and the mixing is **adiabatic** what is the final temperature of the solution.

PROBLEM 4. EXCESS ENTALPY MODEL for ISOPROPANOL-WATER MIXTURE

A. For the narrow temperature range (0°C to 50°C) where the experimental data were taken thermodynamically consistent Excess property functions are given by polynomials of the form: $M^E = x_1 x_2 (A_0 + A_1 x_1 + A_2 x_1^2 + \cdots)$ where M is the thermodynamic property in our case Enthalpy. We assume that terms $A_3 x_1^3 + \cdots$ do not contribute significantly so the model for excess enthalpy can be written

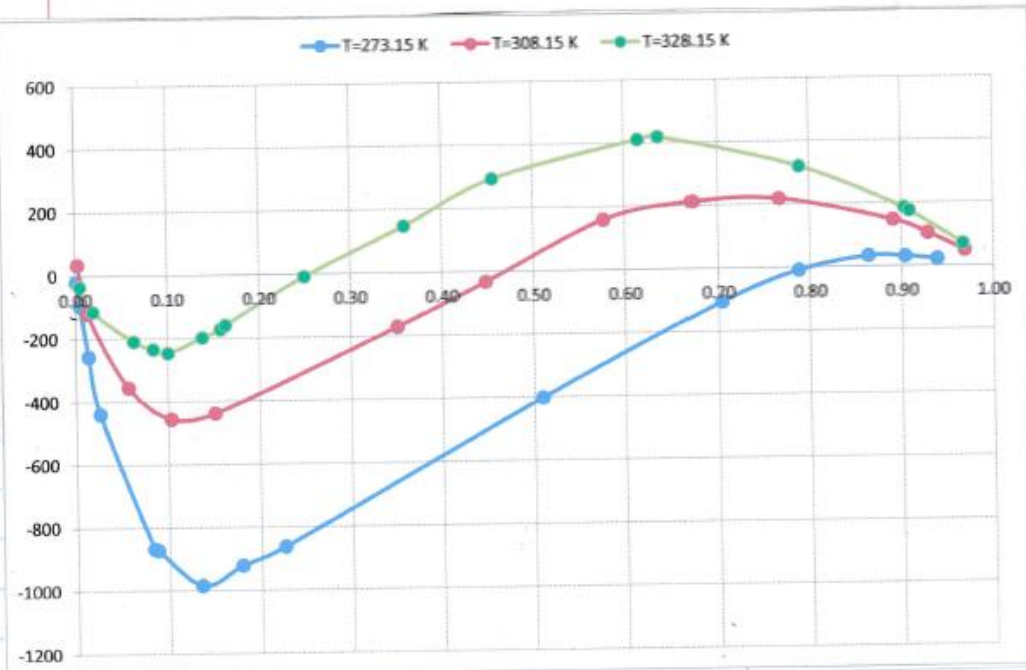
$$H^E = x_1 x_2 (A_0 + A_1 x_1 + A_2 x_1^2)$$ where A_0, A_1, A_2 constants which are functions of temperature must be determined from the experimental data

Performing multiple-linear Regression in Excel we get

T	A_0	A_1	A_2
273.15	-11900.20	30664.97	-19744.7
308.15	-6330.68	19380.44	-12324.6
328.15	-3572.27	15183.79	-10306.3

To identify the effect of temperature, we observe that if we plot A_0, A_1, A_2 vs. ln T (or also vs T) we obtain straight lines with very high linear correlation $R^2 > 0.98$ for all coefficients ▬

Below we see a graph of the excess enthalpy model against the exxperimental data at all three temperature levels.



The linear functional dependence of the coefficients A_0, A_1, A_2 along with the final Excess Enthalpy Model are summarized here

$A_0 = 45503 \ln T - 267150$ $T [=] K$

$A_1 = -85617 \ln T + 510{,}734$

$A_2 = 52796 \ln T - 315662$

$H^E = x_1 x_2 (A_0 + A_1 x_1^2 + A_2)$ $[= kJ/kmol]$

B. Now that we developed the model for Excess enthalpy we can calculate the molar enthalpy of any mixture using the formula

$$H = x_1 H_1 + x_2 H_2 + H^E \Rightarrow H = x_1 H_1 + x_2 H_2 + x_1 x_2 (A_0 + A_1 x_1 + A_2 x_1^2)$$

where $H_1(P,T) = H_1^v(P^{sat}, T) - \Delta H_1^{vap}(T)$ and
$H_2(P,T) = H_2^v(P^{sat}, T) - \Delta H_2^{vap}(T)$

where $H_1^v(P_1^{sat}, T)$, $H_2^v(P_2^{sat}, T)$ is the enthalpy of vapor at temperature T and pressure P_i^{sat} where P_i^{sat} is the saturated vapor pressure of component i

$\Delta H_1^{vap}(T)$ and $\Delta H_2^{vap}(T)$ are the heats of vaporization of the two components

The saturated vapor pressures of the two components Isopropyl alcohol & water for temperatures < 50 °C are very low, so we can ignore the effect of pressure on liquid enthalpy and also assume that the vapor @ P^{sat} behaves like an ideal gas.
Based on these assumptions

$$H_1(T) = H_1^{IG}(T) - \Delta H_1^{vap}(T) \qquad (Eq. 1)$$

$$H_2(T) = H_2^{IG}(T) - \Delta H_2^{vap}(T) \qquad (Eq. 2)$$

Let's now calculate total enthalpy before and after mixing. To do that we can we our calculator and (Eq. 1) & (Eq. 2) in page (5).

BEFORE MIXING $H_1^{tot} = n_1 H_1$ where $n_1 = 20$

$H_2^{tot} = n_2 H_2$ $n_2 = 80$

AFTER MIXING $H^{tot} = n(x_1 H_1 + x_2 H_2 + H^E)$ $n = n_1 + n_2 = 100$

The heat released from the mixing @ 315.15 °K is then given by $Q = H^{tot} - H_1^{tot} - H_2^{tot} \Rightarrow$

$$Q = n(x_1 H_1 + x_2 H_2 + H^E) - n_1 H_1 - n_2 H_2 \Rightarrow \boxed{Q = n H^E}$$

To calculate H^E @ 315.15 °K first we calculate

$A_0 = 45503 \ln(315.15) - 267150 \Rightarrow A_0 = -5369.0$

$A_1 = -85617 \ln(315.15) + 510{,}734 \Rightarrow A_1 = 18175.2$

$A_2 = 52796 \ln(315.15) - 315{,}662 \Rightarrow A_2 = -11924.0$

Using these values of obtain $H^E = -353.8$ kJ/kmol and therefore the heat released

$$Q = n H^E = 0.1 \cdot (-353.8) \Rightarrow$$

$$\boxed{Q = -35.38 \text{ kJ}}$$

C. For adiabatic mixing we have that the total enthalpy before is the same as the total enthalpy after the mixing.

Let's assume that the temperature after the mixing is equal to Tm, and before the mixing is T. Then from the energy balance we have

$$n_1 H_1(T) + n_2 H_2(T) = n\left(x_1 H_1(T_m) + x_2 H_2(T_m) + x_1 x_2 \left(A_0(T_m) + A_1(T_m)x_1 + A_2(T_m)x_1^2\right)\right)$$

This equation must be solved for Tm. Dividing left and right hand side by n we obtain

$$x_1 H_1(T) + x_2 H_2(T) = x_1 H_1(T_m) + x_2 H_2(T_m) + x_1 x_2 \left(A_0(T_m) + A_1(T_m)x_1 + A_2(T_m)x_1^2\right)$$

The extended secant method was used with results shown here

A_0	-4938.6377	x_1	0.2000		
A_1	17365.4263	x_2	0.8000		
A_2	-11424.6731				
Before Mixing		H_1	H_2	H^E	**Total**
Temperature	42.0000	-314520.72	-285142.93	0.0000	-291018.49
After Mixing		H_1	H_2	H^E	**Total**
Temperature	44.9950	-313980.19	-284893.55	-307.6063	-291018.49

The temperature of the solution after the mixing rises by approximately 3°C to around 45°C

Problem 5: VLE Data Reduction with the NRTL Model

The following ***Pxy*** data for a Toluene/n-Decane mixture are available at $T = 373.47\ ^{o}\mathrm{K}$:

TEMPERATURE	PRESSURE	X_1	X_2	Y_1	Y_2
K	Pa	TOLUENE	n-DECANE	TOLUENE	n-DECANE
373.47	9450	0.0000	1.0000	0.0000	1.0000
373.47	11630	0.0353	0.9647	0.2415	0.7585
373.47	14180	0.0700	0.9300	0.3834	0.6166
373.47	17240	0.1223	0.8777	0.5147	0.4853
373.47	22710	0.1880	0.8120	0.6544	0.3456
373.47	29530	0.3006	0.6994	0.7732	0.2268
373.47	34260	0.3665	0.6335	0.8277	0.1723
373.47	45240	0.5301	0.4699	0.9029	0.0971
373.47	50040	0.6144	0.3856	0.9251	0.0749
373.47	58680	0.7769	0.2231	0.9560	0.0440
373.47	66370	0.8617	0.1383	0.9746	0.0254
373.47	75380	1.0000	0.0000	1.0000	0.0000

From the **Gibbs Excess** energy function for the **NRTL** (Non-Random Two Liquid) model the liquid activity coefficients for the two mixture components are given by:

$$\ln\gamma_1 = x_2^2\left[\tau_{21}\left(\frac{G_{21}}{x_1 + x_2 G_{21}}\right)^2 + \frac{\tau_{12}G_{12}}{\left(x_2 + x_1 G_{12}\right)^2}\right]$$

$$\ln\gamma_2 = x_1^2\left[\tau_{12}\left(\frac{G_{12}}{x_2 + x_1 G_{12}}\right)^2 + \frac{\tau_{21}G_{21}}{\left(x_1 + x_2 G_{21}\right)^2}\right]$$

where

$\ln G_{12} = -0.3\tau_{12}$ and $\ln G_{21} = -0.3\tau_{21}$

A. Prove the following functional expressions:

$$\ln\gamma_1^{\infty} = \tau_{21} + \tau_{12}\exp\left(-0.3\tau_{12}\right)$$
$$\ln\gamma_2^{\infty} = \tau_{12} + \tau_{21}\exp\left(-0.3\tau_{21}\right)$$

B. Determine whether that experimental data are thermodynamically consistent.

C. Estimate from the experimental data the **NRTL** model coefficients $\tau_{12,}G_{12}$ and $\tau_{21,}G_{21}$ and draw a parity plot of predicted pressure values vs. experimental values. (*Hint: Use the expressions for liquid activity coefficients at infinite dilution to obtain initial guesses for the constants*)

PROBLEM 5. VLE Data Reduction with NRTL

A. $\ln \gamma_1^{\infty} = \ln \gamma_1 (x_1 \to 0, x_2 \to 1)$

From the given NRTL equation for $\ln \gamma_1$ we obtain by substituting $x_1 = 0$ and $x_2 = 1$

$$\ln \gamma_1^{\infty} = \tau_{21} + \tau_{12} G_{12} \Rightarrow \ln \gamma_1^{\infty} = \tau_{21} + \tau_{12} \exp(-0.3\, \tau_{12})$$

In similar manner

$$\ln \gamma_2^{\infty} = \ln \gamma_2 (x_2 \to 0, x_1 \to 1) \Rightarrow$$

$$\ln \gamma_2^{\infty} = \tau_{12} + \tau_{21} G_{21} \Rightarrow \ln \gamma_2^{\infty} = \tau_{12} + \tau_{21} \exp(-0.3\, \tau_{21})$$

B. To determine whether the experimental data are thermodynamically consistent we need to estimate the integral

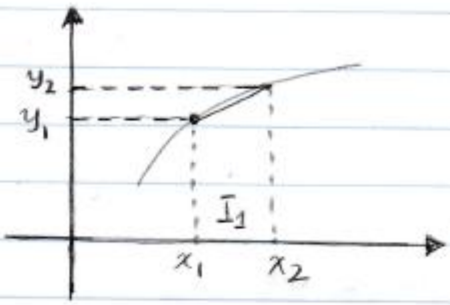
$$I = \int_0^1 \ln\left(\frac{\gamma_1}{\gamma_2}\right) dx_1$$

First we notice that $K_1 = \frac{y_1}{x_1} = \gamma_1 \frac{P_1^{sat}}{P} \Rightarrow$

$$\gamma_1 = \frac{y_1}{x_1} \frac{P}{P_1^{sat}} \quad \text{and also} \quad \gamma_2 = \frac{y_2}{x_2} \frac{P_2^{sat}}{P}$$

From these last two equations we can calculate the value of $\ln \gamma_1 - \ln \gamma_2 = \ln\left(\frac{\gamma_1}{\gamma_2}\right)$

The integral $\int_0^1 \ln\left(\frac{\gamma_1}{\gamma_2}\right) dx_1$ can be calculated using the trapezoid rule. Looking at the graph below the trapezoid rule is given by



$$I_1 = \frac{y_1 + y_2}{2}(x_2 - x_1)$$

In order to calculate the integral we need values of γ_1^∞ and γ_2^∞. To find these values we perform a linear extrapolation from the two closest points

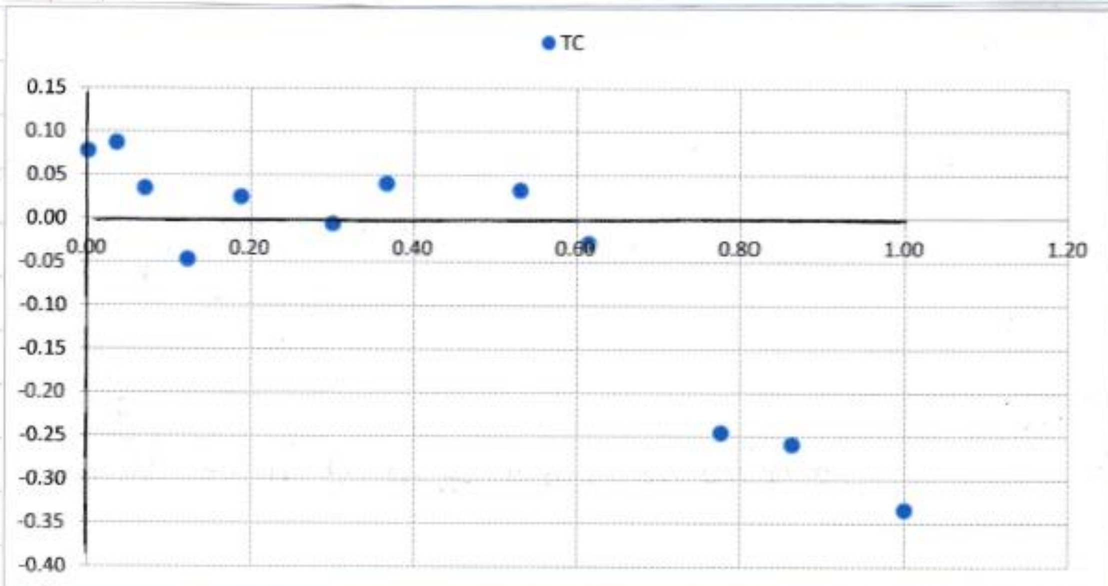
$$\gamma_1^\infty \cong 1.0555 + \frac{1.0555 - 1.0303}{0.0353 - 0.0700}(0 - 0.0353) \Rightarrow \gamma_1^\infty \approx 1.0811$$

$$\gamma_2^\infty = 1.2899 + \frac{1.2899 - 1.2246}{0.1383 - 0.2231}(0 - 0.1383) \Rightarrow \gamma_2^\infty \approx 1.3963$$

Better values of the liquid activity coefficients at infinite dilution can be obtain by consider quadratic approximations.

Now that we know the values of γ_1 and γ_2 for the entire range of $x_1 \in [0, 1]$ we can calculate the integral using the trapezoid rule. The numerical results as well the plot of $\ln(\gamma_1/\gamma_2)$ vs. x_1 are shown below:

Plot of $\ln(\gamma_1/\gamma_2)$ vs x_1



Calculation of the integral under the curve

γ_1	γ_2	$\ln(\gamma_1) - \ln(\gamma_2)$	x_1	Section Integral
1.0811	1.0000	0.0780	0.0000	0.002912
1.0555	0.9676	0.0869	0.0353	0.002116
1.0303	0.9949	0.0350	0.0700	-0.000310
0.9625	1.0087	-0.0469	0.1223	-0.000720
1.0487	1.0228	0.0250	0.1880	0.001090
1.0077	1.0133	-0.0056	0.3006	0.001138
1.0264	0.9860	0.0401	0.3665	0.005967
1.0222	0.9892	0.0328	0.5301	0.000176
0.9995	1.0286	-0.0286	0.6144	-0.022285
0.9579	1.2246	-0.2457	0.7769	-0.021386
0.9958	1.2899	-0.2587	0.8617	-0.040975
1.0000	1.3963	-0.3338	1.0000	-0.072278

The value of the integral $J = -0.07$ appears to indicate that the data are thermodynamically consistent

C. Now we must calculate the coefficients τ_{12}, G_{12}, τ_{21} and G_{21} for the NRTL model. Using the values of the liquid activity coefficients at infinite dilution along with the formulas we developed in Part A. of this problem we obtain

$$\ln \gamma_1^{\infty} = \tau_{21} + \tau_{12} \exp(-0.3\tau_{12}) = 0.078$$

$$\ln \gamma_2^{\infty} = \tau_{12} + \tau_{21} \exp(-0.3\tau_{21}) = 0.334$$

Solving these two equations with the Excel Solver we obtain

$$\tau_{12} = 1.4450 \Rightarrow G_{12} = 4.242$$
$$\tau_{21} = -0.8587 \Rightarrow G_{21} = 0.4237$$

and since $\ln G_{12} = A_{12} + B_{12}/T$
$\ln G_{21} = A_{21} + B_{21}/T$

we may set $A_{12} = 1.445 \quad B_{12} = 0$
$A_{21} = -0.8587 \quad B_{21} = 0$

We can now perform a non-linear regression using the model $\boxed{P = x_1\gamma_1 P_1^{sat} + x_2\gamma_2 P_2^{sat}}$

where γ_1, γ_2 are estimated using the NRTL model.
Numerical results and values of the constants A_{12}, A_{21} are shown

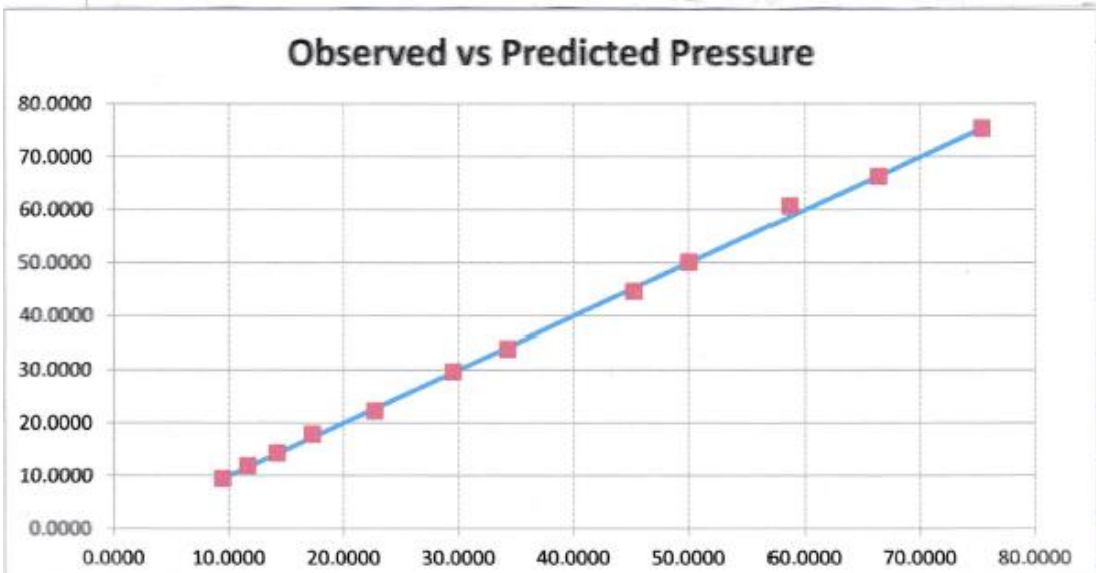
Non-Linear Regression with the NRTL model

Experimental			NRTL			
γ_1	γ_2	$\ln(\gamma_1) - \ln(\gamma_2)$	γ_1	γ_2	Predicted Pressure (kPa)	$(Residual)^2$
1.0607	1.0000	0.0589	1.0607	1.0000	9.4500	0.0000
1.0555	0.9676	0.0869	1.0523	1.0001	11.9177	0.0828
1.0303	0.9949	0.0350	1.0450	1.0005	14.3071	0.0161
0.9625	1.0087	-0.0469	1.0355	1.0015	17.8529	0.3757
1.0487	1.0228	0.0250	1.0259	1.0032	22.2363	0.2244
1.0077	1.0133	-0.0056	1.0141	1.0069	29.6343	0.0109
1.0264	0.9860	0.0401	1.0094	1.0092	33.9290	0.1095
1.0222	0.9892	0.0328	1.0026	1.0146	44.5691	0.4501
0.9995	1.0286	-0.0286	1.0010	1.0168	50.0648	0.0006
0.9579	1.2246	-0.2457	0.9999	1.0191	60.7058	4.1039
0.9958	1.2899	-0.2587	0.9999	1.0192	66.2797	0.0081
1.0000	1.0172	-0.0171	1.0000	1.0172	75.3800	0.0000

LEAST-SQUARES SOLVER PANEL

SSQ	5.38227
A12	-0.58924
A21	0.76206
B12	0.00000
B21	0.00000
C	0.30000

Parity Plot of Predicted Pressure vs. Observed



Notice that the initial guesses for A_{12} & A_{21} are very bad because the calculation of γ_1^∞ and γ_2^∞ was not very good.